Соответственно обе наши оценки и смещенная и несмещенная несостоятельны.

в) $\tilde{\Theta}_3 = x_{max} = \Phi(x)$

$x_{max} \sim (F(x))^n$;

• Несмещенность:

$$Mx_{max} = \int_0^\Theta x \cdot d\Phi(x) = \int_0^\Theta x \cdot n (F(x))^{n-1} \cdot p\, dx = \int_0^\Theta x \cdot n \cdot \frac{1}{\Theta} \cdot \left(\frac{x}{\Theta}\right)^{n-1} dx =$$

$$\cancel{\frac{n}{\Theta^2}\int_0^\Theta x^2 dx = \frac{n}{\Theta^2}} = \int_0^\Theta \frac{x^n}{\Theta^n} \cdot n\, dx = \frac{x^{n+1}}{n+1} \cdot \frac{n}{\Theta^n}\Big|_0^\Theta = \frac{\Theta n}{n+1} \neq \Theta$$

Соответственно оценка смещена

• Несмещённая оценка: $\frac{x_{max}}{n} \cdot (n+1) = \tilde{\Theta}_3'$

• Состоятельность:

$$\cancel{Mx_{max}} \cdot D\left(\frac{x_{max}}{n}(n+1)\right) = \frac{(n+1)^2}{n^2} \cdot Dx_{max} = \frac{(n+1)^2}{n^2}\left(Mx_{max}^2 - \underbrace{\left(Mx_{max}\right)^2}_{\frac{\Theta^2 n^2}{(n+1)^2}}\right) =$$

$$Mx_{max}^2 = \int_0^\Theta x^2 \cdot n \left(\frac{x}{\Theta}\right)^{n-1} \cdot \frac{1}{\Theta} dx = \int_0^\Theta \frac{x^{n+1}}{\Theta^n} \cdot n\, dx = \frac{n}{\Theta^n} \frac{x^{n+2}}{n+2}\Big|_0^\Theta =$$

$$= \frac{n}{\Theta^n} \cdot \frac{\Theta^{n+2}}{n+2} = \frac{n}{n+2} \cdot \Theta^2;$$

$$D\left(\frac{x_{max}}{n}(n+1)\right) = \left(\frac{n}{n+2}\Theta^2 - \frac{\Theta^2 n^2}{(n+1)^2}\right)\frac{(n+1)^2}{n^2} = \frac{(n+1)^2}{(n+2)n}\Theta^2 - \Theta^2 =$$

$$= \Theta^2\left(\frac{\cancel{n^2}+\cancel{2n}+1-\cancel{n^2}-\cancel{2n}}{(n+2)n}\right) = \frac{\Theta^2}{(n+2)n};$$

$\frac{\Theta^2}{(n+2)n} \xrightarrow[n\to\infty]{} 0 \Rightarrow$ по достаточному условию оценка состоятельна

• Способ через определение:

$$\forall \varepsilon > 0: \ P(|\tilde{\Theta}_3' - \Theta| \geq \varepsilon) \xrightarrow[n\to\infty]{} 0$$

1) $P(\tilde{\Theta}_3' \geq \Theta + \varepsilon) = P\left(\frac{x_{max}}{n}(n+1) \geq \Theta + \varepsilon\right) =$

$$= 1 - P\left(x_{max} < \frac{(\Theta+\varepsilon)\cdot n}{(n+1)}\right) = \cancel{1 - P\left(\frac{(\Theta+\varepsilon)\cdot n}{(n+1)}\right) =}$$

$$\cancel{= 1 - \left(F\left(\frac{(\Theta+\varepsilon)n}{(n+1)}\right)\right)^n = 1 - \left(F\left(\frac{(\Theta+\varepsilon)n}{(n+1)\Theta}\right)\right)^n = 1 -}$$

$$\cancel{= \left(\frac{\Theta n}{\Theta(n+1)} + \frac{\varepsilon n}{(n+1)\Theta}\right)^n} - \left(\frac{\Theta+\varepsilon}{\Theta}\frac{n}{n+1}\right)^n =$$

$\Big\{ x_{max}$ точно меньше либо равен Θ,